

▪ AOF - Append only file

- This writes all write operations to the log file as only append files but not rewrite files.
- AOF file is essentially a redo log, through which the database state can be restored.
- When the data is restored, the file is read to rebuild the data, that is, if the database is restarted. The *write* command is executed from front to back according to the content of the log file to complete the data recovery work.
- As the number of commands executed increases, the size of the AOF file increases, which can lead to several problems, such as increased disk usage and slow restart loading. But there are several ways to control this issue.

The BASE database model

In a distributed environment, the BASE concept is generally applied, prioritising availability over consistency. The BASE model emphasises high availability as the primary principle.

NoSQL, in general, follows BASE. NoSQL databases, which store data in documents such as key-value pair, wide-column, pure documents, and graphs rather than the traditional relational model, are easy to query. NoSQL enables rapid changes, believes in zero downtime, and supports large users distributed globally.

While NoSQL databases usually adhere to BASE principles, some of them also hold certain degrees of ACID compliance.

Given below are the three main properties of BASE.

Basic availability

The system must guarantee availability to all users at all times, even in the event of failures. Ideally, the system should always be available, reach the replica in case of failures, not lock up, and the data should be available while the operation is in progress. Immediate consistency is not the priority; the key is assurance of availability.

Soft-state

As we are not consistent, the stored values may change due to the eventual consistency model. Here we could get stale information which was in sync before. It becomes the developer's responsibility to ensure consistency.

Eventual consistency

Currently the data may be stale, but the system state is gradually replicated to all the nodes. Data reads are still possible, but may not reflect reality.

From the above principles it's quite clear that BASE favours availability over consistency.

The CAP theorem

Now let's look into the CAP theorem; it is the key to understand when your database needs to be distributed. It can be applied to both NoSQL and SQL databases.

Some systems need to be live all the time — 24*7*365. However, it could be tricky to have them running in a single server and single data centre, as there could be hardware crashes, power failure or any natural disaster in the city. Consequently, we always need to anticipate and plan for system failures and design our system to run from different servers and different locations at all times to improve user experiences. Additionally, data should be reliable and highly available.

The CAP theorem is also called Brewer's Theorem, because it was first put forth by Professor Eric A. Brewer during a talk he gave on distributed computing in 2000. Two years later, MIT professors Seth Gilbert and Nancy Lynch published a proof of 'Brewer's Conjecture'.

Consistency

This consistency is different from the ACID model's consistency. Here, a consistent view of data at each node means all the nodes always have the same and most recent view of the data. If not the most recent or if it is not consistent, an error must be returned. Any write to the database should be immediately replicated to the replicas. This will be challenging since syncing globally at the same time is hard. There are several techniques such as change log capturing, querying based on the time stamps, acknowledgement from maximum nodes, and so on, that provide the fastest replication across the nodes.

Also, at times, consistency goes for a toss when we read from the non-updated cache. An example of consistency

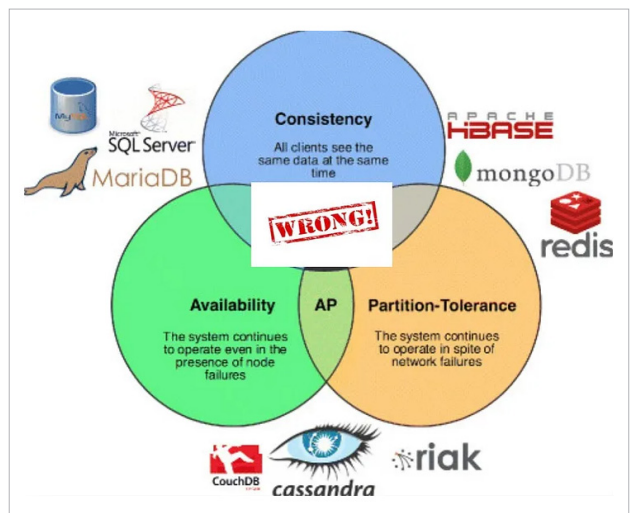


Figure 2: CAP (Credit: <https://medium.com/@sumitsethia94/consistency-or-availability-of-databases-do-you-really-understand-cap-8ecf2b3bb099>)